

Logs Workshop

Log Management & Analytics at scale!

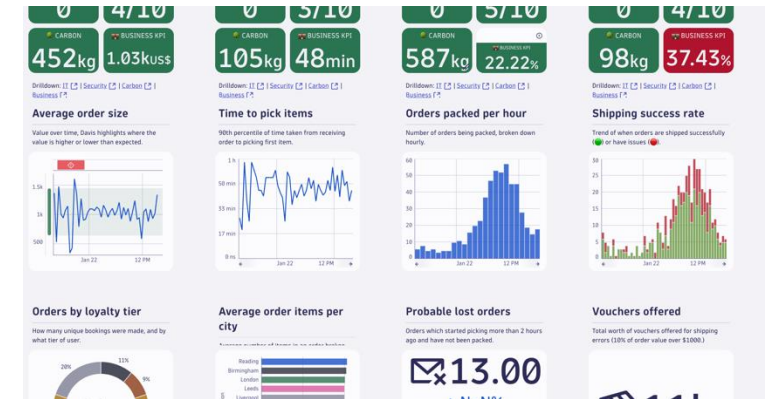
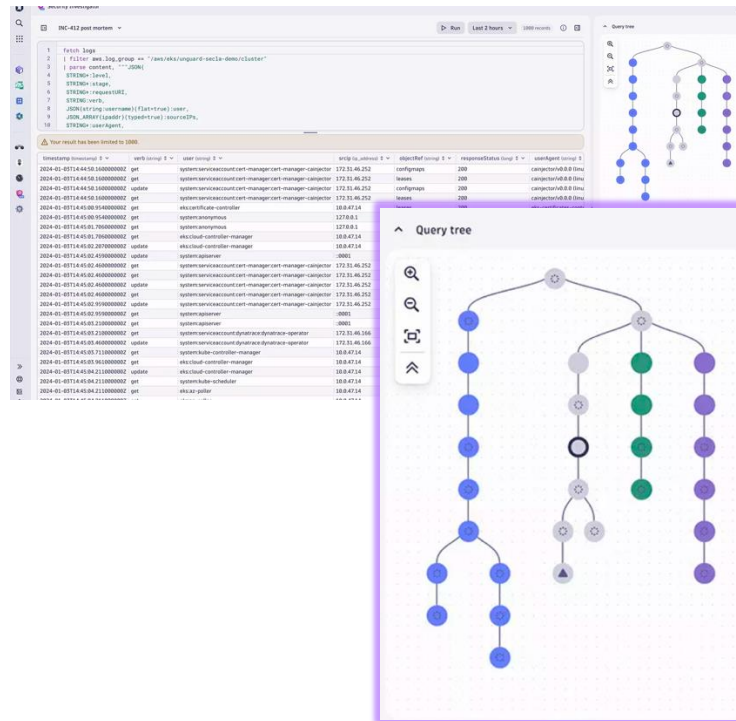
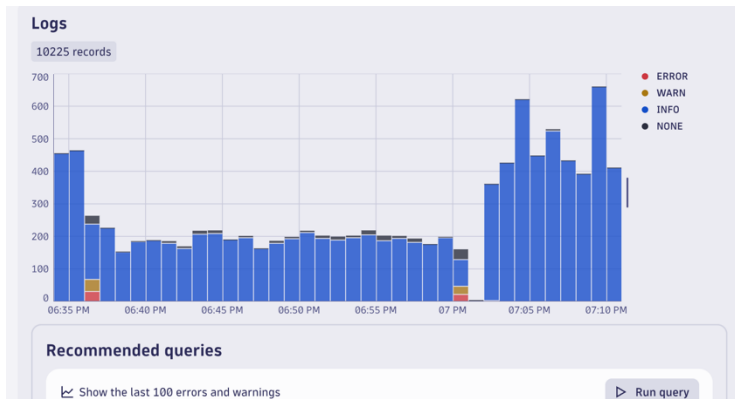
Agenda

- Log Management and Analytics in Dynatrace provides instant answers to critical questions across:
 - Business
 - DevOps
 - SRE
 - Operations
 - Security
- This technical session will cover best practices and strategies for Log Management and Analytics in Dynatrace, including scalable implementation.
- Attendees will gain knowledge on managing enterprise logging in Dynatrace and leave with actionable insights for their own implementations.
- **Part 1: Persona – Dynatrace Users**
 - Log Management and Analytics
 - Hands-on Training
 - Logs Explorer
 - DQL Training
 - Logs to Metrics
 - Dashboarding
- **Part 2: Persona - Dynatrace Admins**
 - Anomaly Detection
 - Buckets and Segments
 - Permissions
 - OpenPipeline
 - Live Demo & Hands-on

Part 1

Persona – All Dynatrace Users

Logs are a key data source for Observability, Security and Business Goals



Analytics, AI, and Automation for Unified Observability and Security



Infrastructure
Observability



Application
Observability



Digital
Experience



Log
Analytics



Application
Security



Threat
Observability



Software
Delivery



Business
Analytics

 dynatrace



AutomationEngine



AppEngine



Smartscape®



Davis® AI



Grail™



Hub



OpenPipeline™



PurePath®



OneAgent®



Topology



Traces



Metrics



Logs



Behavior



Code



Metadata



Network



Security Events



Threats

Discussion Question

What are necessary capabilities for a Log Management and Analytics Solution?

Ingest logs from many sources

Retain logs for selected timeframe

Permissions and Access Controls

Discussion Question

Scalability

What are necessary capabilities for a Log Management and Analytics Solution?

Filtering

Alerts from log data

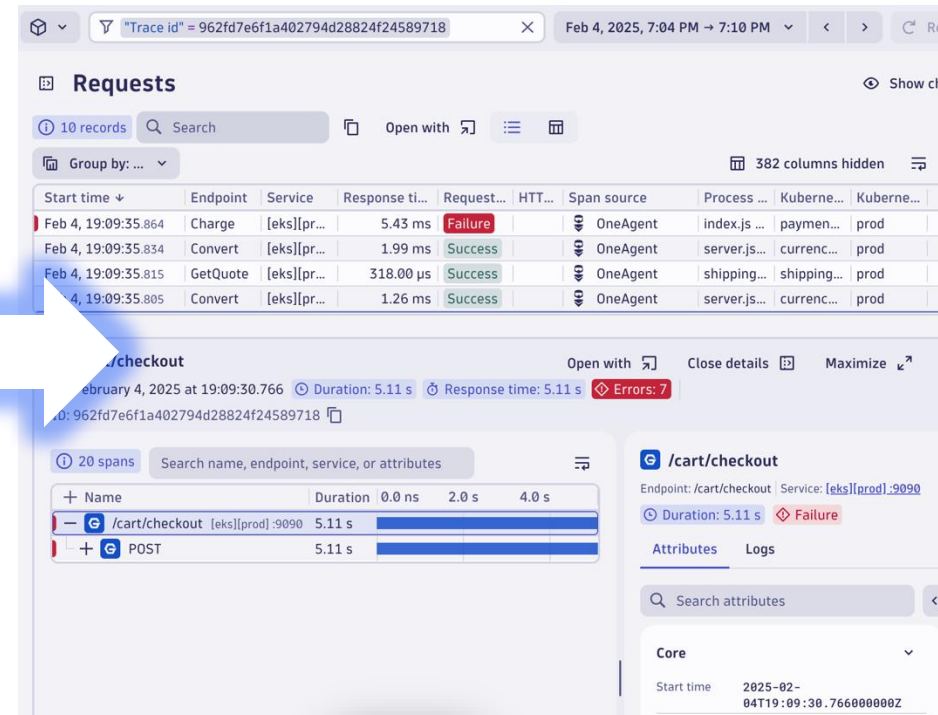
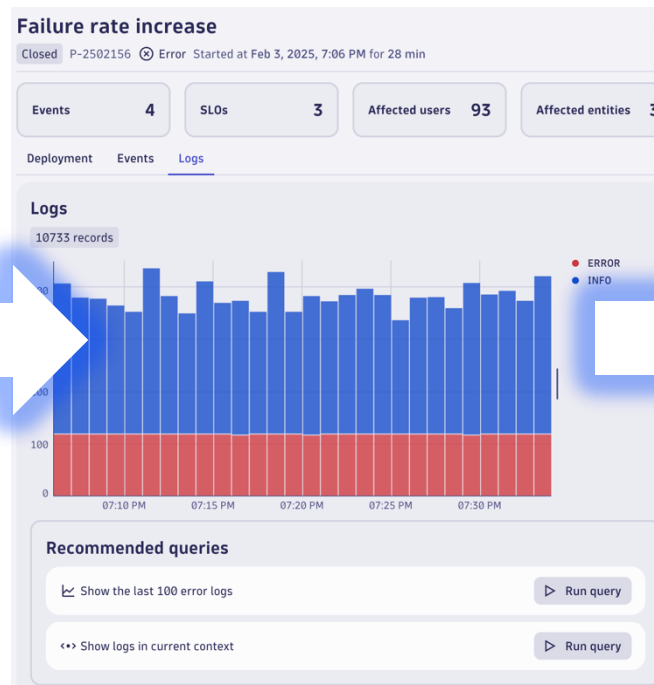
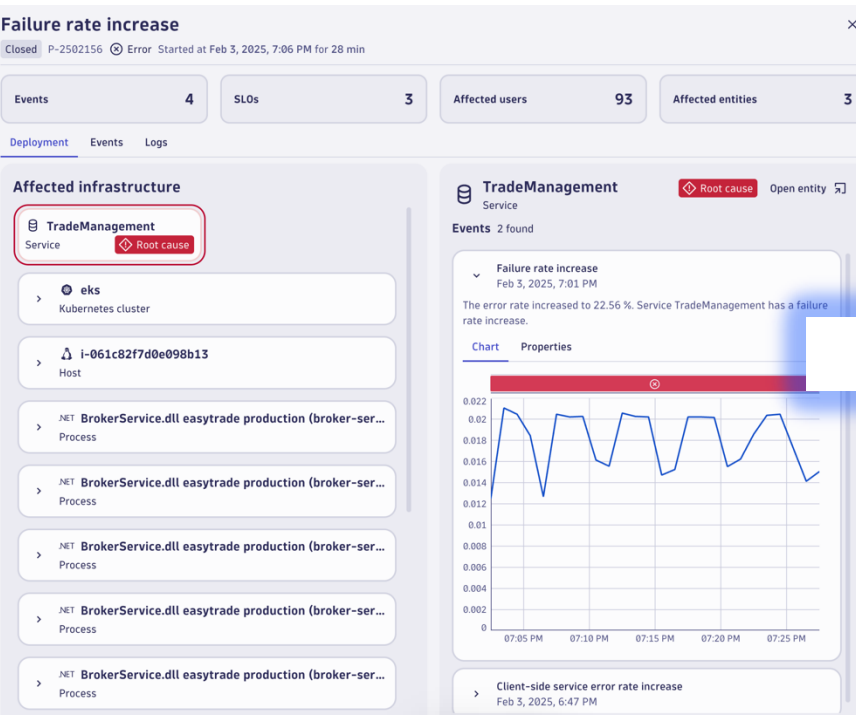
Automatic parsing for standard logs

Deep dive analytics

Data security

Easily get answers from Logs!

Logs in context reduces MTTR and provides instant answers



We used to spend hours manually searching through metrics, logs, and traces to piece together insights about user experience. Now, this takes minutes or seconds.

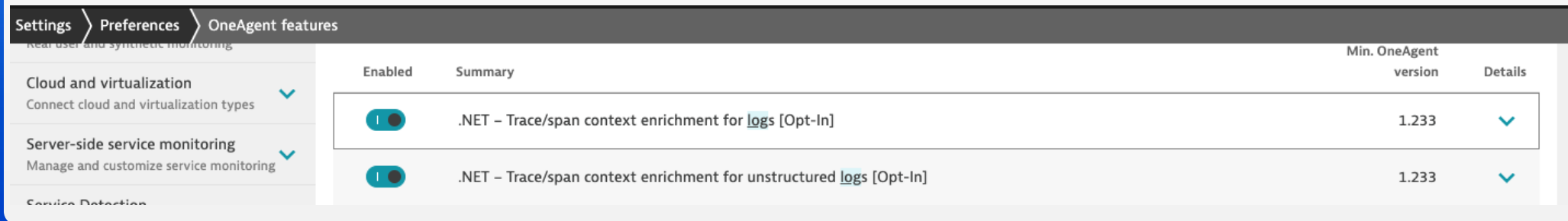


Discussion Question

How does Dynatrace enrich logs with trace context?

Discussion Question

How does Dynatrace enrich logs with trace context?



The screenshot shows the 'OneAgent features' settings page in Dynatrace. On the left, there is a sidebar with navigation links: 'Settings', 'Preferences', and 'OneAgent features'. Below these, there are three expandable sections: 'Real user and synthetic monitoring', 'Cloud and virtualization', and 'Server-side service monitoring'. The main content area displays a table of features. The table has four columns: 'Enabled', 'Summary', 'Min. OneAgent version', and 'Details'. Two features are listed, both with their 'Enabled' toggle switches turned on. The first feature is '.NET - Trace/span context enrichment for logs [Opt-In]' with a minimum version of 1.233. The second feature is '.NET - Trace/span context enrichment for unstructured logs [Opt-In]' also with a minimum version of 1.233.

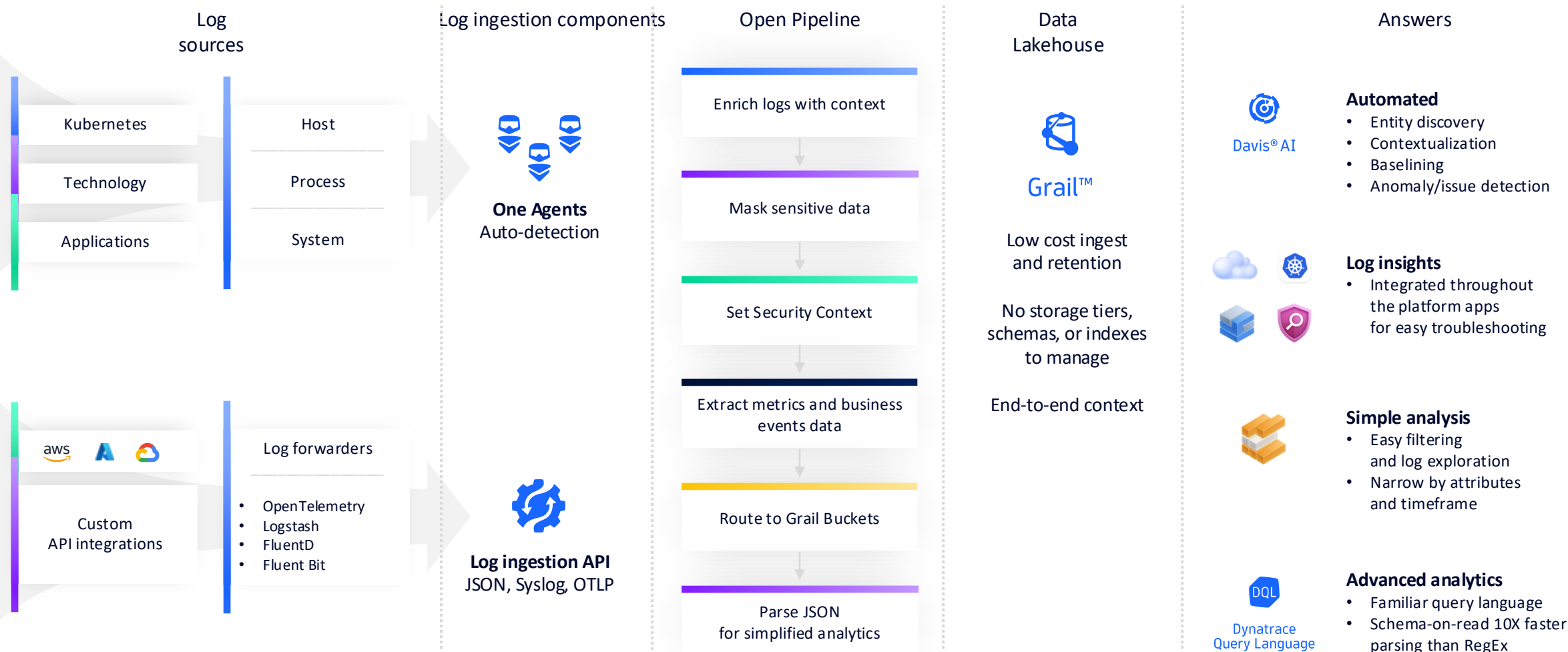
Enabled	Summary	Min. OneAgent version	Details
☒	.NET - Trace/span context enrichment for logs [Opt-In]	1.233	▼
☒	.NET - Trace/span context enrichment for unstructured logs [Opt-In]	1.233	▼

Automatically for popular logging frameworks!
Just toggle it on in OneAgent Features

Dynatrace also provides guides for enrichment for
other logging methods or for use with
OpenTelemetry

Log Management & Analytics workflow

Fast insights without the painful data management



Discussion Question

What is a bucket in Dynatrace?

Buckets are a way to organize data in Dynatrace.

Buckets can be used for data retention, separating use cases, and access permissions.

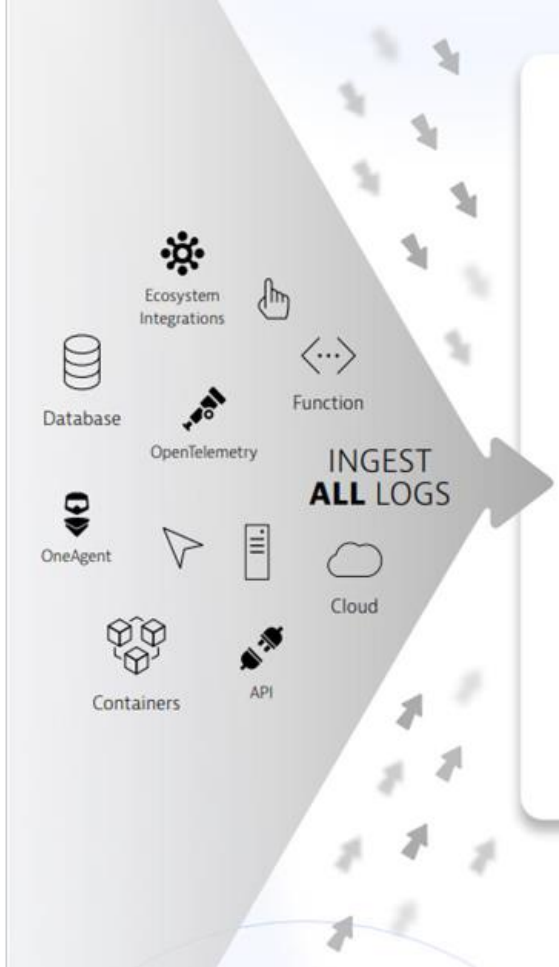
Discussion Question

What is a bucket in Dynatrace?

Buckets can be used to improve query performance by reducing query execution time and the scope of data read.

Scaling Log Analytics

Scaling Log Analytics



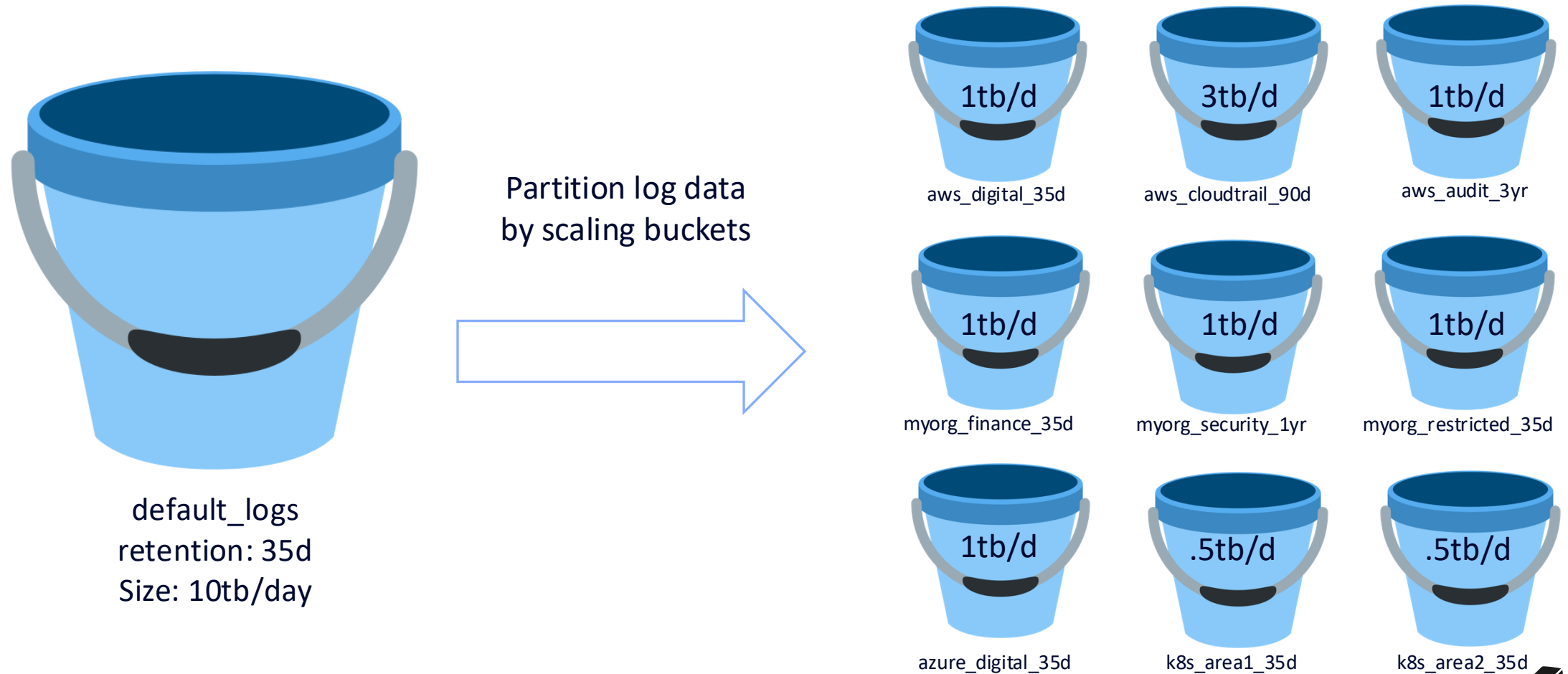
default_logs
retention: 35d
Size: 10tb/day

```
1 fetch logs
2 | filter isnotnull(k8s.deployment.name)
3 | summarize Count = count(), : {k8s.deployment.name}
4 | sort Count desc
5 | limit 10
6
```

10 rec... Executed at: 1/12/2025, 22:09:23, Timeframe: 1/5/2025, 22:09:21 - 1/12/2025, 22:09:21, Scanned bytes: 500 GB

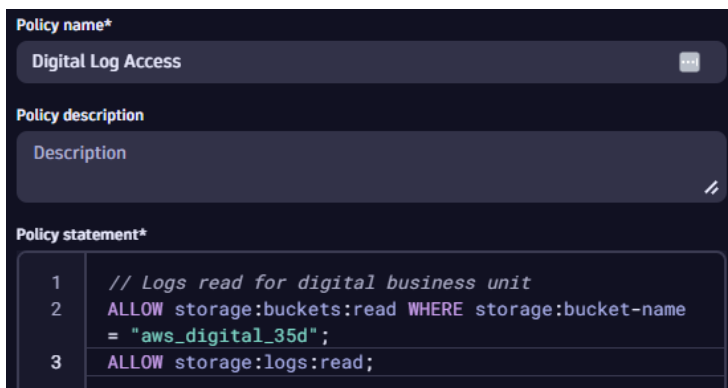
k8s.deployment.name	Count
unguard-malicious-load-generator-*	39,054,339
unguard-user-simulator-*	31,541,074
offerservice-*	14,599,336
calculationservice-*	13,817,824
ingress-nginx-controller-*	13,233,959
pvc-out-of-space-kaniko-big-image-push-*	10,944,380
currencyserviceproxy-*	9,513,482
frontendreverseproxy-*	5,521,804
astroshop-imageprovider-*	5,030,592
aggregator-service-*	4,461,697

Scaling Log Analytics

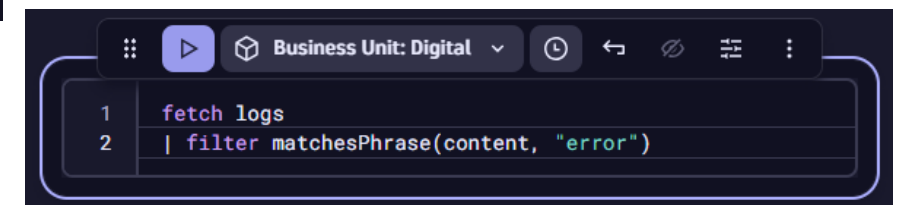
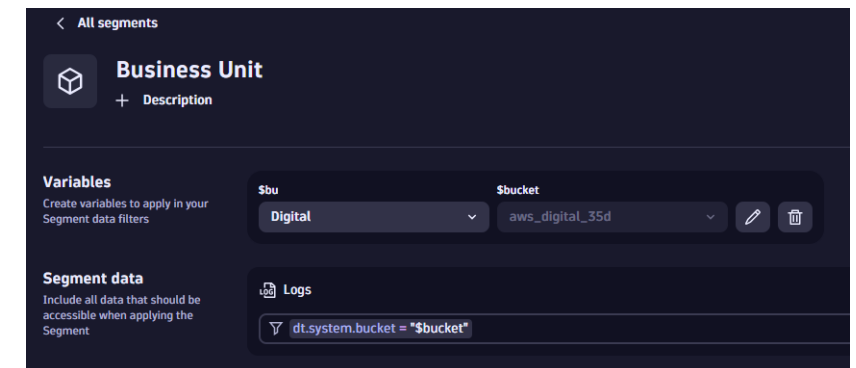


Understanding Query Behavior

- DQL and in app queries will scan all buckets a user has permission to read by default.
- Dynatrace automatically optimizes queries based on filters
- To realize performance and cost optimizations from a bucket strategy, queries must be filtered to relevant buckets.
 - This can be accomplished in three ways:
 - Only give users access to relevant buckets
 - Apply dt.system.bucket filter into DQL query
 - Utilize segments for targeting appropriate buckets



```
1 fetch logs
2 // only return logs that are stored in the following bucket
3 | filter dt.system.bucket == "k8s_area1_35d"
4
5 // can also use matcher functions such as matches phrase:
6 // | filter matchesPhrase(dt.system.bucket, "k8s")
```



Segments

Why Segments?

Settings

- Monitoring
Setup and overview
- Cloud Automation
Setup and configuration
- Processes and containers
Monitoring, detection and naming
- Web and mobile monitoring
Real user and synthetic monitoring
- Cloud and virtualization
Connect cloud and virtualization types
- Server-side service monitoring
Manage and customize service monitoring
- Service Detection
Define rules for services and spans
- Log Monitoring
Set up management of logs
- Anomaly detection
Configure detection sensitivity
- Alerting
Configure alerting settings
- Dashboards
Configure dashboard settings

Management zones settings

Management zones enable defining fine grained access rights to parts of an environment. A Management zone consists of a set of entities like applications, hosts, process groups, or services.

 Your user does not have the necessary write permissions.

Filter items...	
Summary	Details
FinOps – Host aks-default-77091117-vmss00001Y TEST	
mz-az-af	
mz-az-af-abssahara	
mz-az-agcs	
mz-az-agcs-cl	
mz-az-agcs-fa	
mz-az-agcs-ipp	
mz-az-agcs-middleware	

Page Unresponsive

You can wait for it to become responsive or exit the page.

 Management zones settings - Environment - Settings - Allianz ASI...

Wait

Exit page

prod
Allianz ASIC

Filter

Select management zone

All

FinOps – Host aks-default-77091117-vmss00001Y TEST

mz-az-af

mz-az-af-abssahara

mz-az-agcs

mz-az-agcs-cl

mz-az-agcs-fa

mz-az-agcs-ipp

mz-az-agcs-middleware

mz-az-agcs-ms

mz-az-agcs-pa

mz-az-agcs-ra

mz-az-agcs-sd

mz-az-agcs-uw

mz-az-agcs-wp

mz-az-at

mz-az-at-appdev

mz-az-at-appdev_playground

mz-az-at-bmproduct

mz-az-au

mz-az-au-aaldeops

mz-az-au-prod

mz-az-ay

mz-az-ay-digitalplatform

mz-az-azp

mz-az-azp-AZGACanadaIT

mz-az-azp-development

mz-az-bcm_de

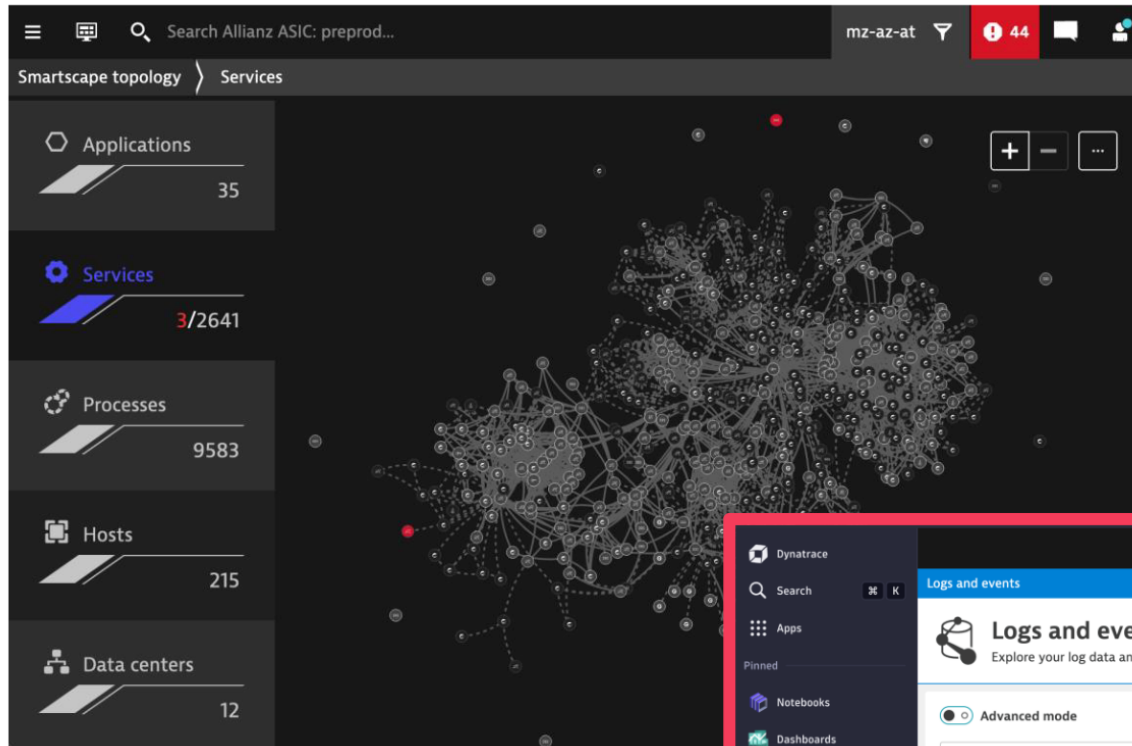
mz-az-bcm_de-dbitkvi5co

mz-az-bcm_de-global

mz-az-bg

mz-az-bg-AZBG_DynaTrace

Why Segments?



Experience the
full power of
 Grail and the
Dynatrace Query
Language

 Where MZ?

Dynatrace

Search [K]

Apps

Pinned

- Notebooks
- Dashboards
- Workflows
- Hub
- Dashboards classic
- Problems

Recently opened

- AWS
- Hosts
- Logs and events

Logs and events

Logs and events Powered by Grail™

Explore your log data and Kubernetes events in simple mode. For deeper analysis, or to query business events, switch to advanced mode powered by DQL

Advanced mode

Filter by

Run query

Search attributes...

Search results Notification Execution time: 2 s Scanned data: 423 MB

Attribute counts are estimated based on sampled data.

Open with... Create processing rule Create metric Format table Actions

timestamp	status	content
2023-11-16 13:21:50.401	NONE	SNMP trap (CISCO-SMI::ciscoMgmt.41.2.0.1) reported from 10.69.0.18
2023-11-16 13:21:50.401	NONE	SNMP trap (CISCO-SMI::ciscoMgmt.41.2.0.1) reported from 10.69.0.19
2023-11-16 13:21:50.401	NONE	{\"timestamp\": \"2023-11-16T12:21:45.191917656Z\", \"message\": \"Update loop too

Segments: Preconfigured Dynamic Filtering

Kubernetes

Explorer

Anomaly detectors

Cluster: CL-Prod20

Filter by: + Add filter

Clusters

Nodes

Namespaces

Workloads

Deployments

DaemonSets

StatefulSets

ReplicaSets

Replication controllers

CronJobs

Jobs

Pods

Services

Containers

Clusters

12 records

Davis® AI

Health status

Clusters

1 / 16

Nodes

4 / 456

Namespaces

137

Workloads

149 / 5.9k

Health

Utilization

Metadata

12 columns hidden

Cluster	Problems	Nodes	Namespaces	Workloads	Pods	CPU Usage	CPU Requests	CPU
code-matrix-cluster	2	12	8	789	1.2k			
quantum-compute-pod	1	1 / 2	5	12	78			
binary-nexus	-	30	12	13 / 568	234			
devops-grid	-	1	5	456	48			
cyberspace-core	-	2 / 500	23	1.3k	89			
robo-mesh-cluster	-	4	3	8 / 90	5			
nano-dene-forge	-	12	1	4 / 45	67			
crypto-mesh-hub	-	1 / 45	24	456	124			
data-pipeline-grid	-	5	4	545	78			
quantum-logic-nest	-	1	3	3	3			

1 click to block out all the noise

Segments Implementation

Trigger

Trigger

We offer two predefined variants:

- Default, where we show the currently applied segment if one segment is selected. Multiple segments are grouped to make the width more predictable.
- Compact, where we only show the amount of selected segments.

Additionally, the trigger is customizable, as for all our overlays.

	No segment	Simple segment	Segment with variable (1 value)	Segment with variable (2+ values)	2+ segments
Default trigger	 ▾	 Shop ▾	 Region: eu-ireland ▾	 Region (2) ▾	 2 segments ▾
Compact trigger	 ▾	 1 ▾	 1 ▾	 1 ▾	 2 ▾

Simple Segment



Show me entities, logs, events, etc. of my application

- Equivalent to single MZ

Multiple Segments

 2 segments ^

Segments

Region	▼	eu-ireland, eu-austria	▼	×
App-env	▼	production	▼	×

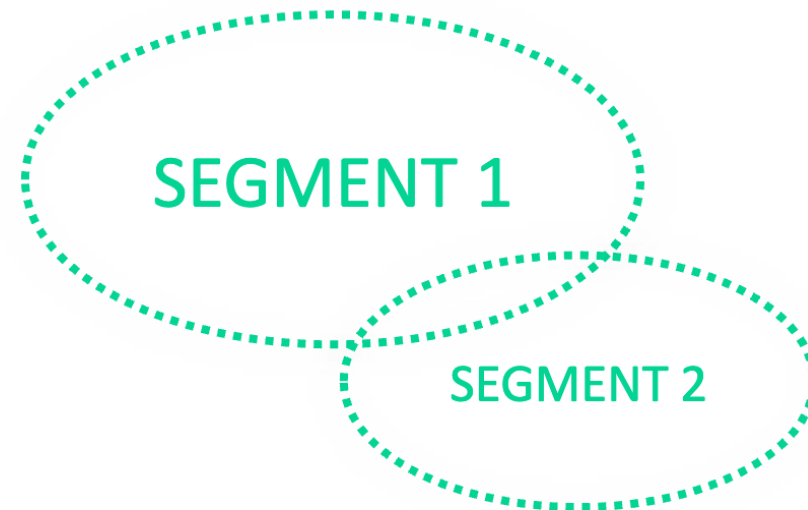
+ Segment

Clear all

[Manage segments](#)[Learn more](#) 

Show me data of my application from a specific infrastructure

- Intersection of multiple segments (eu-ireland OR eu-austria) AND production



How it all works

 2 segments ^

Segments

Region

eu-ireland, eu-austria

×

App-env

production

×

+ Segment

Clear all

[Manage segments](#)[Learn more](#)



1) Segment IDs and values are passed to query API

FiRmGMvWzPY(eu-ireland, eu-austria), HCkccp8aAIU(production)

2) Grail filters consumer queries with conditions defined in segments

```
fetch logs
| filter
  // permission-based filters
  ... AND
  // Segment-based filters
  (
    (condition_a1 OR condition_a2 OR ...) AND
    (condition_b1 OR condition_b2 OR ...)
  )
  // consumer query continues
| summarize ...
```

Transparent
filter conditions

Segments and in-app filters

Kubernetes Explorer Recommendations

2 Additional filter for further problem analysis

Region (2) Type to filter Update

Filter by segments

Region eu-ireland, eu-austria

1 Segments selector to set context

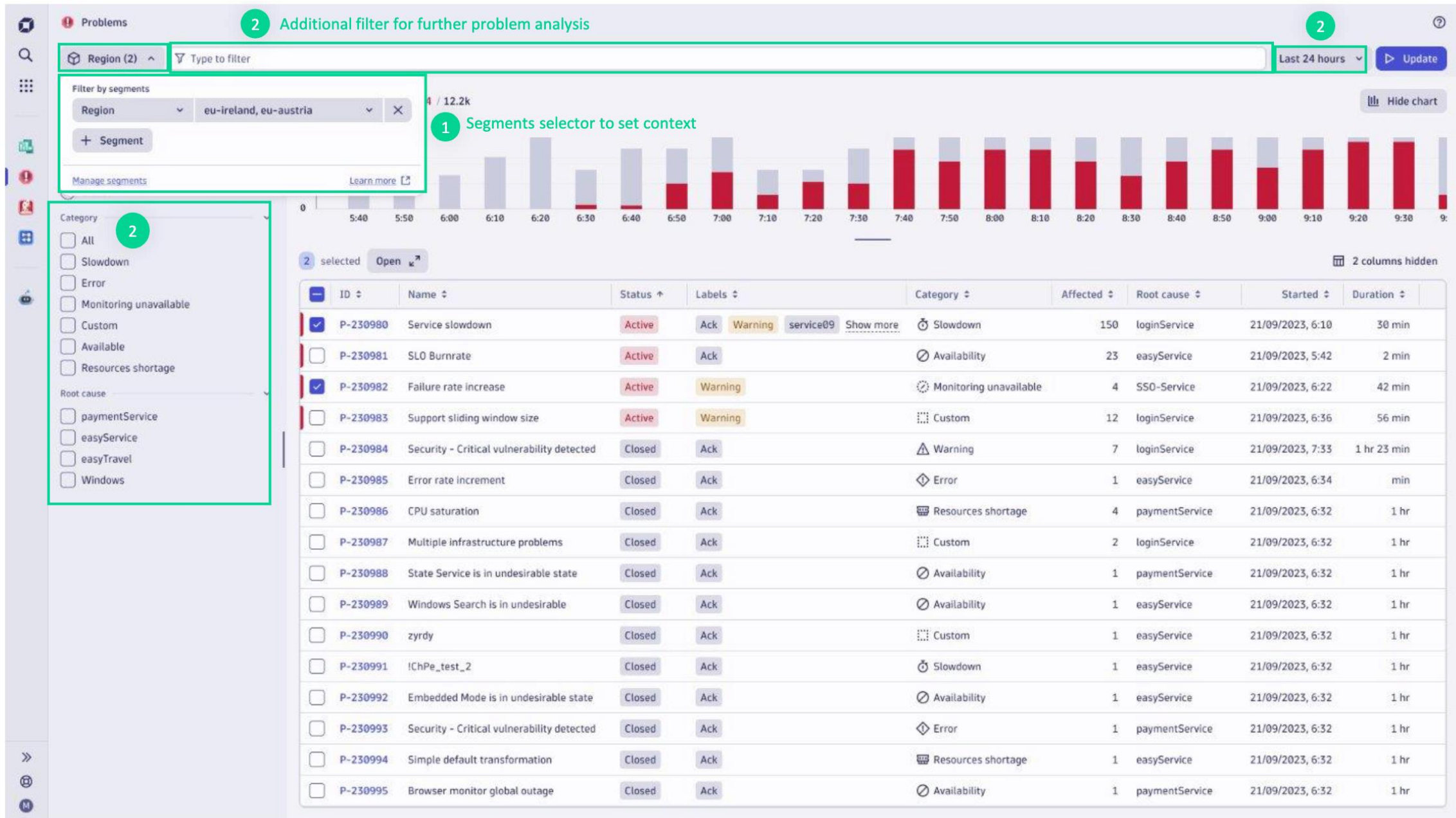
24 Nodes 3 / 3 Namespaces 12 Workloads 3.5k / 12.7k

Problems Topology 12 columns hidden

Node	Problems	Pods	Workloads	Services	Usage	Requests	alertmanager	apiserver	key	key
full-observability...	-	123	3	45			main	main	✓	-
full-observability...	-	12	3	345			longvaluetextexample...	✓	-	-
GKE CP KLU	-	3	12	3			main	main	✓	-
OCP 3.11	-	12	456	1			-	longvaluetextexample...	✓	-
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
ocp4-9-in-cluster	-	4	89	3			longvaluetext	longvaluetextexample...	-	-
ocp4-9-kuberne...	-	4	45	54			main	longvaluetext	-	-
rauter-cri-test	-	45	456	34			✓	✓	longvaluetextexample...	main
trauter-e2e	-	5	545	1,234			-	-	✓	longvaluetext e
full-observability...	-	123	3	45			main	main	✓	-
full-observability...	-	12	3	345			longvaluetextexample...	✓	-	-
GKE CP KLU	-	3	12	3			main	main	✓	-
OCP 3.11	-	12	456	1			-	longvaluetextexample...	✓	-
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
ocp4-9-in-cluster	-	4	89	3			longvaluetext	longvaluetextexample...	-	-
ocp4-9-kuberne...	-	4	45	54			main	longvaluetext	-	-
rauter-cri-test	-	45	456	34			✓	✓	longvaluetextexample...	main
trauter-e2e	-	5	545	1,234			-	-	✓	longvaluetext e
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
trauter-e2e	-	1	3	3			-	✓	longvaluetextexample...	-

20 rows per page

Page 1 of 2



Building a Segment

< All segments

This segment has unsaved changes.

Discard changes

Save

ACE DT

(Full-stack) Observability data for ACE DT team.

Owner

R Roman Windischhofer

Visibility

Anyone in the environment

2 Variables for dynamic segments

Variables

Create variables to apply in your Segment data filters

\$namespace.name

dynatrace

\$namespace.id

CLOUD_APPLICATION_NAM...

3 Include data directly

Logs

k8s.namespace.name = \$namespace.name

Preview

Metrics

k8s.namespace.name = \$namespace.name

Preview

4 Include data of entities

Hosts

tags = team:ACE

Preview

Include

Problems

Vulnerabilities

Metrics

Logs

Events

BizEvents

Spans

5 Include data of related entities

Services

runs_on = \$this.hosts

Preview

Include

Problems

Vulnerabilities

Metrics

Logs

Events

BizEvents

Spans

1 Visible to anyone or unlisted

20 include blocks per segment

1 filter for data of entities (logs, events, spans)

No limit for problems, vulnerabilities, metrics

1 topology traversal step only

+ All types + Metrics + Logs + Events + BizEvents + Spans + Entities

12

Discussion Question

Why would you want to create a metric from a log?

Query Performance

Cost efficient storage and query

Discussion Question

Why would you want to create a metric from a log?

Simplified alert creation



Why Logs to Metrics?



- **Enhanced Performance:**

- Efficient storage
- Faster query performance
- Better scalability for handling of large volumes of data

- **Simplified Alerting:**

- Easier to setup and manage Metrics-based alerts
- Easier to generate dynamic thresholds
- Easy to aggregate and summarize data

- A metric can be either a Counter or Value type.

- Counter metric - # of HTTP requests

- Value metric - Product quantity, Revenue

The screenshot displays the 'Metric Extraction' configuration interface. On the left, a list of metrics is shown with toggle switches: 'getCart' (Counter metric), 'Slow cart' (Counter metric), 'failPayments' (Counter metric), 'order Checkout' (Counter metric), 'successful payments' (Counter metric), 'Procesed orders' (Counter metric), 'Received Requests' (Counter metric), 'Shipped Requests' (Counter metric), 'Revenue' (Value metric), and 'shipping' (Counter metric). The 'getCart' and 'Revenue' metrics are highlighted with red boxes. To the right, the configuration details for these metrics are shown. For 'getCart', the 'Name*' is 'getCart', the 'Matching condition*' is 'matchesPhrase(content, "GetCartAsync")', and the 'Metric key*' is 'log. getCart' (indicated by a red arrow). For 'Revenue', the 'Name*' is 'Revenue', the 'Matching condition*' is 'matchesPhrase(k8s.deployment.name, "paymentservice-*") and matchesPhrase(Message, "Transaction processed:")', the 'Field extraction*' is 'Amount', and the 'Metric key*' is 'log. revenue' (indicated by a red arrow). Both metrics have 'Pre-defined' dimensions selected.





Log to Events

1/1/2025 12:00:00 AM

An event can be either Davis or Business event:



Davis Event

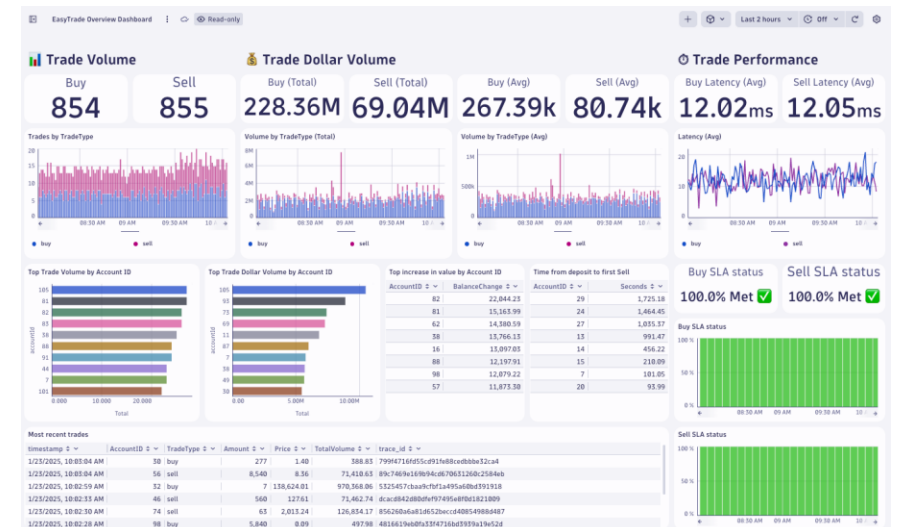
- Does this anomaly need to be reported?
- Is it an INFO Event that needs to be generated?



Business Event

- Is this data for reporting business data?
- Is this data for analytics for your Ops team?

Problem Analysis - Last 24 Hours							
Problems							
Status	Problem	Affected	StartTime	EndTime	Duration	eventid	
OPEN	P-25013783 - High CPU throttling	unguard-user-simulator	1/17/2025, 5:59:00 PM	In Progress	5.67 d	-1021638725722261559_1737154740000V2	
OPEN	P-25015137 - Job failure event	pvc-out-of-space-kaniko-big-image-push	1/23/2025, 1:01:00 AM	In Progress	9.06 h	29026935580201881192_1737612060000V2	
CLOSED	P-25015026 - Job failure event	pvc-out-of-space-kaniko-big-image-push	1/22/2025, 5:01:00 PM		3.25 h	6507152318141236067_1737581260000V2	
CLOSED	P-25014994 - Job failure event	pvc-out-of-space-kaniko-big-image-push	1/22/2025, 12:46:00 PM		4.25 h	5751904292146087284_1737567960000V2	
CLOSED	P-25014950 - Job failure event	pvc-out-of-space-kaniko-big-image-push	1/22/2025, 7:01:00 AM		5.75 h	-5931107189908644196_1737547260000V2	
CLOSED	P-25015101 - Out-of-memory kills	paymentservice	1/22/2025, 10:22:00 PM		15.00 min	3345252367425892024_1737602520000V2	
CLOSED	P-25015021 - Memory usage close to limits	paymentservice	1/22/2025, 4:52:00 PM		5.73 h	-6865210298140368054_1737582720000V2	
CLOSED	P-25014976 - Memory usage close to limits	manager	1/22/2025, 9:45:00 AM		1.03 h	-1432535056653467008_1737557100000V2	
OPEN	P-25015212 - Not all pods ready	mail-service	1/23/2025, 9:30:00 AM	In Progress	34.58 min	-1194887507921013297_1737642600000V2	
CLOSED	P-25015211 - Pods stuck in pending	mail-service	1/23/2025, 9:10:00 AM		45.00 min	4536517296561041882_1737641400000V2	
CLOSED	P-25014971 - Not all pods ready	mail-service	1/22/2025, 9:30:00 AM		50.00 min	-2528180704712407912_1737556200000V2	
CLOSED	P-25015175 - Container restarts	ingress-dev-controller	1/23/2025, 5:04:00 AM		16.00 min	-592571486552286267_1737626640000V2	





Dashboarding

- Spot the differences between these two dashboards?



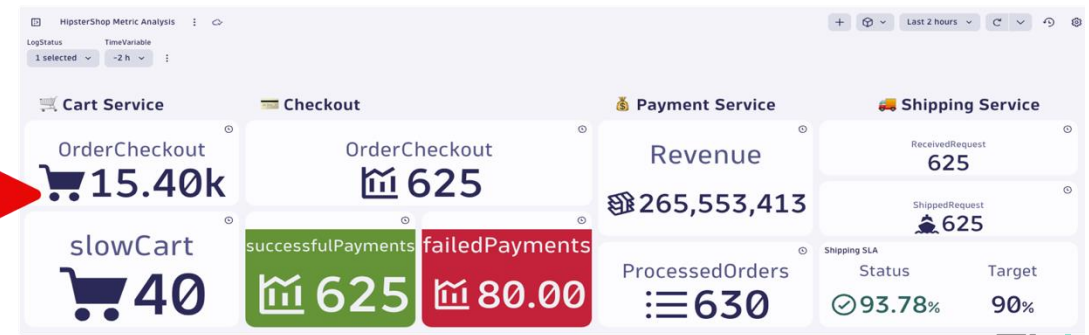
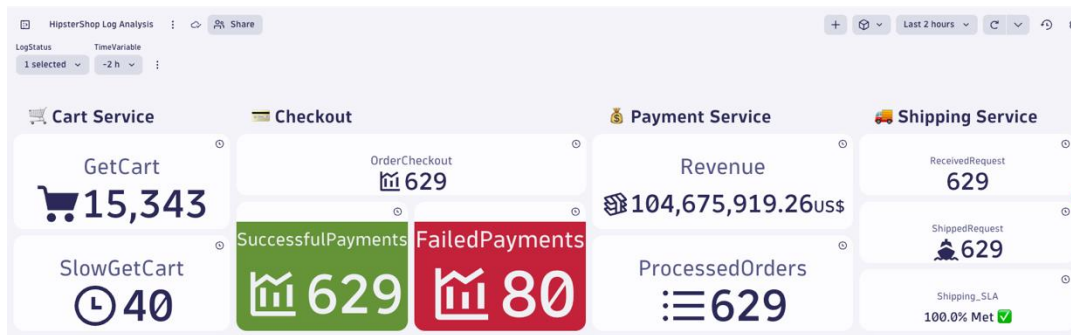
Use logs when:

Low frequency refresh
Involve a complex DQL
Joining different types of data
Analyzing short timeframes



Use metrics when:

High frequency refresh
Trending over long timeframes (timeseries)
Optimize query cost
To create SLOs alerts





Alerting



• Use logs when:

- Infrequent ingest of log records
- Complex query
- Detailed investigations



• Use metrics when:

- Frequent data points
- Dynamic thresholds
- Performance tracking



Training Environment

1. Log out of any existing Dynatrace environments.
2. Login to the following tenant:

<https://<XXXXXX>.sprint.apps.dynatracelabs.com/>

ID: XXXX@protonmail.com

Password: Dynatrace123*

Hands-on Training

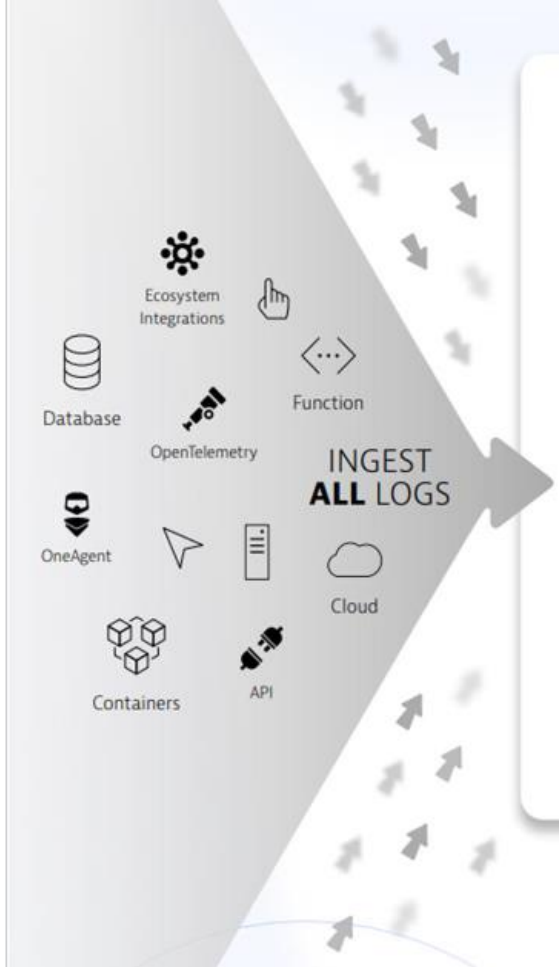
End of Part 1

Part 2

Persona – Dynatrace Admins

Scaling Log Analytics

Scaling Log Analytics



default_logs
retention: 35d
Size: 10tb/day

```
1 fetch logs
2 | filter isnotnull(k8s.deployment.name)
3 | summarize Count = count(), : {k8s.deployment.name}
4 | sort Count desc
5 | limit 10
6
```

10 rec... Executed at: 1/12/2025, 22:09:23, Timeframe: 1/5/2025, 22:09:21 - 1/12/2025, 22:09:21, Scanned bytes: 500 GB

k8s.deployment.name	Count
unguard-malicious-load-generator-*	39,054,339
unguard-user-simulator-*	31,541,074
offerservice-*	14,599,336
calculationservice-*	13,817,824
ingress-nginx-controller-*	13,233,959
pvc-out-of-space-kaniko-big-image-push-*	10,944,380
currencyserviceproxy-*	9,513,482
frontendreverseproxy-*	5,521,804
astroshop-imageprovider-*	5,030,592
aggregator-service-*	4,461,697

Scaling Log Analytics



default_logs
retention: 35d
Size: 10tb/day

Partition log data
by scaling buckets



1tb/d

aws_digital_35d



3tb/d

aws_cloudtrail_90d



1tb/d

aws_audit_3yr



1tb/d

myorg_finance_35d



1tb/d

myorg_security_1yr



1tb/d

myorg_restricted_35d



1tb/d

azure_digital_35d



.5tb/d

k8s_area1_35d



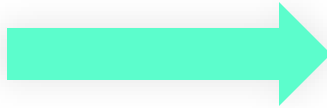
.5tb/d

k8s_area2_35d



Define your bucket scaling strategy

- Developing a tailored data partitioning (bucket) strategy should gather the following information:
 - Ingest sources
 - Estimated Ingest volume
 - Retention desires
 - Compliance Requirements (audit)
 - Sensitive data & masking
 - Permission requirements



Bucket strategy best practice:

- Define a strict naming standard such as <provider>_<type>_<retention> or <provider>_<bu>_retention.
- Consider Business units or App Groupings
- Dedicate buckets to your largest ingest sources.
- Ingest of 1tb per day is ideal.
- Additional permissions such as restricted access can be built on top of the standard:
aws_hr_restricted_35d &
aws_hr_nonrestricted_35d

Additional bucket considerations

- 80 bucket limit per tenant by default
 - Can be significantly increased based on ingest volume – contact support
- Buckets exist for all data types in Grail
 - Logs, events, BizEvents, spans, RUM, Metrics etc..
- Strategy can be replicated for other high volume data types such as BizEvents or spans.

- More buckets = more setup / admin management – strike a balance
- Ensure bucket design aligns with available attributes on log records (host group, account / Sub-IDs, log source)
- Entity properties are not available for lookups to map records to buckets.

How to create buckets

- Storage Management App or Platform API (/platform/storage/management/v1)

< Buckets

↻

+ Bucket

You can create, update, or delete buckets here. Explore our documentation for [Assigning Access Rights](#) or [Data Removal Procedures](#).

Filter

Select bucket type

Search buckets

logs

Condensed

Default

Comfortable

4 columns hidden

Bucket name	Display name	Retention period (days)	Table	Status	Actions
aws_audit_3yr	AWS Audit 3yr	1,095	logs	Active	
aws_cloudtrail_90d	AWS CloudTrail 90d	90	logs	Active	
aws_digital_35d	AWS Digital 35d	35	logs	Active	
azure_digital_35d	Azure Digital BU 35d	35	logs	Active	
default_logs	Logs	35	logs	Active	
k8s_area1_35d	K8s Area 1 35d	35	logs	Active	
k8s_area2_35d	K8s Area 2 35d	35	logs	Active	
myorg_finance_35d	Finance logs 35d	35	logs	Active	
myorg_restricted_35d	Restricted access logs 35d	35	logs	Active	
myorg_security_1yr	Security Logs 1yr	365	logs	Active	



Permissions

- Controlling access to logs can be accomplished via IAM policies and defining bucket level and/or record level access.
- Bucket level control is done via bucket names.
- Record level control is done via fields or `dt.security.context` which must be set on the records themselves and defined within the policy.

All bucket access with deny

Policy name*

Generic Log Viewer

Policy description

Description

Policy statement*

```
1 // Logs read all except restricted
2 ALLOW storage:buckets:read WHERE storage:table-name
  = "logs";
3 DENY storage:buckets:read WHERE storage:bucket-name
  = "myorg_restricted_35d";
4 ALLOW storage:logs:read;
```

Digital bucket read & security context = hipstershop

Policy name*

Hipster App Team Logs

Policy description

Description

Policy statement*

```
1 // Allow digital bucket read but restrict to
  specific records
2 ALLOW storage:buckets:read WHERE storage:bucket-name
  = "aws_digital_35d";
3 ALLOW storage:logs:read WHERE
  storage:dt.security_context = "hipstershop";
4
```



Permissions: Supported Fields

Field name	IAM condition	Supported IAM tables
<code>event.kind</code>	<code>storage:event.kind</code>	<code>events</code> , <code>bizevents</code> , <code>system</code>
<code>event.type</code>	<code>storage:event.type</code>	<code>events</code> , <code>bizevents</code> , <code>system</code>
<code>event.provider</code>	<code>storage:event.provider</code>	<code>events</code> , <code>bizevents</code> , <code>system</code>
<code>k8s.namespace.name</code>	<code>storage:k8s.namespace.name</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code> , <code>spans</code>
<code>k8s.cluster.name</code>	<code>storage:k8s.cluster.name</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code> , <code>spans</code>
<code>host.name</code>	<code>storage:host.name</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code> , <code>spans</code>
<code>dt.host_group.id</code>	<code>storage:dt.host_group.id</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code> , <code>spans</code>
<code>metric.key</code>	<code>storage:metric.key</code>	<code>metrics</code>
<code>log.source</code>	<code>storage:log.source</code>	<code>logs</code>
<code>dt.security_context</code>	<code>storage:dt.security_context</code>	<code>events</code> , <code>bizevents</code> , <code>system</code> , <code>logs</code> , <code>metrics</code> , <code>spans</code> , <code>entities</code>
<code>gcp.project.id</code>	<code>storage:gcp.project.id</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code>
<code>aws.account.id</code>	<code>storage:aws.account.id</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code>
<code>azure.subscription</code>	<code>storage:azure.subscription</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code>
<code>azure.resource.group</code>	<code>storage:azure.resource.group</code>	<code>events</code> , <code>bizevents</code> , <code>logs</code> , <code>metrics</code>



Discussion Question

What is OpenPipeline?

OpenPipeline is the data handling solution (pipeline) for Dynatrace to seamlessly ingest and process data.

Used for configuring processing, transforming, metric or data extraction, permissions and storage.

Discussion Question

What is OpenPipeline?

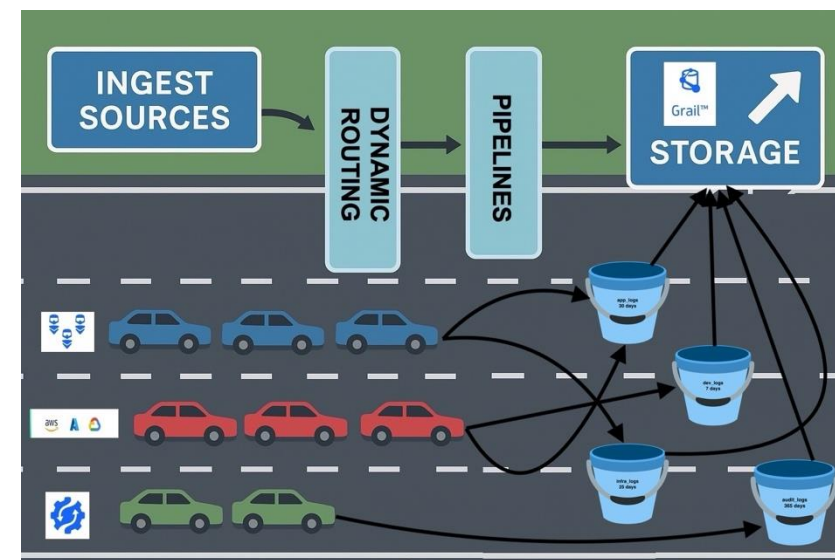
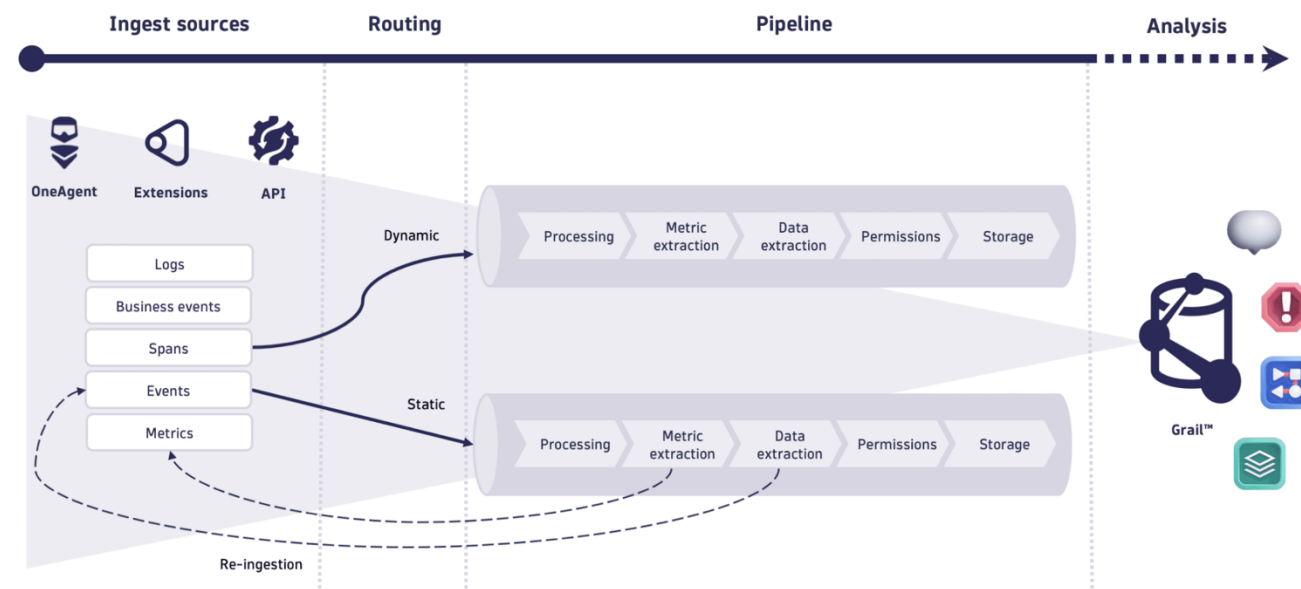
Unified solution for ingestion and processing of different data types, not just logs.



OpenPipeline

- OpenPipeline is used to normalize log records with processing rules, create metrics, create events, assign security context, and define storage to buckets
- Mapping incoming log records to specific buckets is done via OpenPipeline.
- OpenPipeline should be strategized in a similar manner to buckets.
 - 1:1 pipeline to bucket
 - 1:many pipeline to buckets

OpenPipeline How does OpenPipeline work?





OpenPipeline

PIPELINES FOR K8S LOGS

K8s Logs ID: pipeline_K8s_Logs_1199 Discard Save

Processing Metric Extraction Data extraction Permission Storage

Processing
Remove or mask sensitive data, reshape format, parse values.

+ Processor

- shipping service DQL ☒
- checkout/payment se... DQL ☒
- revenue payment serv... DQL ☒

shipping service
DQL ID: processor_payment_service_8218

Name*
shipping service

Matching condition*
1 matchesPhrase(k8s.deployment.name, "shippingservice-*")

DQL processor definition*
1 parse content,"JSON:Data"
2 | fieldsAdd Message=Data[message],Sev=Data[severity]
3 | fieldsAdd slatotal = if(contains(Message,"received request"), "Received"
4

Sample data

```
1 {  
2   "timestamp": "2025-01-16T00:49:23.647434000-06:00",  
3   "content": "{\\\"message\\\":\\\"[GetQuote] received request\\\",  
4   \"Message\": \"[GetQuote] received request\",  
5   \"dt.openpipeline.pipelines\": [  
6     ],  
7   \"logs:pipeline_K8s_Logs_1199\"  
8 }  
9   \"k8s.deployment.name\": \"shippingservice-*\"
```

You can test the selected processor on sample data. Fetch sample data from [Notebooks](#)

Run sample data

- OpenPipeline works in sequential mode, first vertical and then horizontal for each pipeline.
- Processing allows you to parse, add/remove/rename fields and drop record.
- Metric extraction allows you to create counter or value metric.



Training Environment

1. Log out of any existing Dynatrace environments.
2. Login to the following tenant:

<https://XXXXX.sprint.apps.dynatracelabs.com/>

ID: XXXXX@protonmail.com

Password: Dynatrace123*

Live Demo

- Buckets
- Segments
- Permissions
- OpenPipeline

End of Part 2